

## Part A — Git Fundamentals & Advanced Commands

```
[Terminal] A1. Initial Commit on Main
$ git commit -m 'feat: initial commit with README and gitignore'
[main (root-commit) clb2329] feat: initial commit with README and gitignore
2 files changed, 39 insertions(+)
create mode 100644 .gitignore
create mode 100644 README.md
```

```
[Terminal] A3. Git Log Graph Across All Branches
$ git log --oneline --graph --all
* 835b025 feat(pipeline): add dvc.yaml stage definitions
b29666ca feat(eval): add evaluate.py for test metrics and confusion matrix
* 40d4000 feat(arch): add train.py for sequential ANN architecture
* 135e8bd feat(preprocess): add preprocess.py for normalization and validation split
* b9871c2 feat(data): add prepare.py to download and extract Fashion-MNIST
* 4b52dc9 feat(config): add params.yaml for centralized hyperparameters
c1b2329 feat: initial commit with README and gitignore
```

```

[Terminal] A3. Git Log --stat -3 (Commit File Modification Stats)

$ git log --stat -3
commit 835b025f09d1a5eaaa761a9e067b1b3365c7ff89a
Author: Abduljalil Kamaal <@kamaal.com>
Date: Mon Oct 5 19:35:10 2020 +0500
    feat: Add evaluation stage definitions
    dvc.yaml | 43 ++++++
    1 changed, 43 insertions(+), 0 deletions(-)
commit b2966c3a385ed25470b9d1a6948963be316c39047
Author: Abduljalil Kamaal <@kamaal.com>
Date: Mon Oct 5 19:35:00 2020 +0500
    feat: Eval - Add evaluate.py for test metrics and confusion matrix
    src/evaluate.py | 67 ++++++
    1 changed, 67 insertions(+), 0 deletions(-)
commit 9671b30135f8acc05848f8e68229747dbb94c4d85
Author: Abduljalil Kamaal <@kamaal.com>
Date: Mon Oct 5 19:35:00 2020 +0500
    feat: Eval - Add train.py for sequential ANN architecture
    src/train.py | 54 +++++
    1 file changed, 54 insertions(+)

```

### A3. Log Variants: `git log -p -1`

```

commit log -p l
git diff 35b2025f09d1a5eaa7631a6e067b1b365c7ff80a
+info: Mon Oct 5 19:35:01 2026 +0500
+Data: Mon Oct 5 19:35:01 2026 +0500
+diff --realtime --dvcyaml b/dvc.yaml stage definitions
+diff --realtime --dvcyaml b/dvc.yaml
+index 00000000..b47c6e9
+diff --null
+@@ -0,0 +1,3 @@
+stage
+Prepare:
+  cmd: python src/prepare.py
+deps:
+  src/prepare.py
+outs:
+  data/raw
+preprocess:
+  cmd: python src/preprocess.py
+  data/raw
+  src/preprocess.py
+.. [truncated for display] ..

```

### A3. Log Variants: `git log main..dev`

[illegible]

#### A4(a). Diff Variants: `git diff` (Unstaged Working Changes)

```
$ git diff
diff -git a/README.md b/README.md
index 8c36fcf..6c1873a 100644
--- a/README.md
+++ b/README.md
@@ -20,3+20,5 @@ python src/train.py
 python src/evaluate.py

+
+<!-- unstaged changes demo -->
```

#### A4(b). Diff Variants: `git diff --staged`

**[Terminal] A4(b). Git Diff --staged (Staged Changes)**

```
$ git diff --staged
diff --git a/README.md b/README.md
index 8c36f1f..6c1873a 100644
--- a/README.md
+++ b/README.md
@@ -20,3+20,5 @@
 python src/train.py
 python src/evaluate.py

+
+<!-- unstaged changes demo -->
```

#### A4(c). Diff Variants: git diff main..dev (Two-Dot)

[Terminal] A4(c). Git Diff Two-Dot (main..dev)

```
$ git diff main..dev
diff --git a/dvc.yaml b/dvc.yaml
new file mode 100644
index 0000000..b47c6e9
+++ b/dvc.yaml
@@ -0,0 +1,43 @@
+stages:
+  prepare:
+    cmd: python src/prepare.py
+    deps: src/prepare.py
+    outs: data/raw
+  preprocess:
+    cmd: python src/preprocess.py
+    deps: data/raw
+    params:
+      - preprocess.seed
+    outs: preprocess.test_size
+... [truncated for display] ...
```

#### A4(c). Diff Variants: git diff main...dev (Three-Dot)

[Terminal] A4(c). Git Diff Three-Dot (main...dev)

```
$ git diff main...dev
diff --git a/dvc.yaml b/dvc.yaml
new file mode 100644
index 0000000..b47c6e9
+++ b/dvc.yaml
@@ -0,0 +1,43 @@
+stages:
+  prepare:
+    cmd: python src/prepare.py
+    deps: src/prepare.py
+    outs: data/raw
+  preprocess:
+    cmd: python src/preprocess.py
+    deps: data/raw
+    params:
+      - preprocess.seed
+    outs: preprocess.test_size
+... [truncated for display] ...
```

#### A5. Stash Scenario: git stash list

[Terminal] A5. Git Stash List

```
$ git stash list
stash@{0}: WIP on dev: 835b025 feat(pipeline): add dvc.yaml stage definitions
```

#### A5. Stash Scenario: git stash pop

[Terminal] A5. Git Stash Pop (Restoring Working State)

```
$ git stash pop
On branch dev
Changes staged for commit:
  (use "git add -p" to update what will be committed)
  (use "git restore --staged" to discard changes in working directory)
Untracked files:
  (use "git add" to include in what will be committed)
  dvcignore
  Assignment3 Report.pdf
  confusion_matrix.png
  dvc.lock
  metrics.json
no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (571ff29c96e2022e298c7fb1715cefbb3b4a2fd8)
```

## A6. Rebase Scenario: git rebase main & Graph

**[Terminal] A6. Rebase dev onto main & Linear Graph**

```
$ git rebase main; git log --oneline --graph --all
Rebasing (1/6)
Rebasing (2/6)
Rebasing (3/6)
Rebasing (4/6)
Rebasing (5/6)
Rebasing (6/6)
Successfully rebased and updated refs/heads/dev.
+ ade9daa feat(pipeline): add dvc.yaml stage definitions
+ 917251e feat(eval): add evaluate.py for test metrics and confusion matrix
+ 917251e feat(model): add train.py for Sequential ANN architecture
+ 7866e5c feat(preprocess): add preprocess.py for normalization and validation split
+ 9742883 feat(data): add prepare.py to download and extract Fashion MNIST
+ 3ce2231 feat(config): add params.yaml for centralized hyperparameters
+ c49687d Merge branch hotfix into main
|/
+ eedc38b fix(docs): improve README header phrasing
|/
+ clb2329 feat: initial commit with README and gitignore
```

## A7. Reset Scenario: git reset --soft HEAD~1

**[Terminal] A7(a). Git Reset --soft HEAD~1 (Changes Kept Staged)**

```
$ git reset --soft HEAD~1 && git status
On branch scratch
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   scratch_file.txt
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    .dvcignore
    Assignment3_Report.pdf
    confusion_matrix.png
    dvc.lock
    metrics.json
```

## A7. Reset Scenario: git reset --hard HEAD~1

**[Terminal] A7(b). Git Reset --hard HEAD~1 (Changes Fully Discarded)**

```
$ git reset --hard HEAD~1 && git status
HEAD is now at 69f27f3 temp: scratch commit 1
On branch scratch
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    .dvcignore
    Assignment3_Report.pdf
    assignment.pdf
    confusion_matrix.png
    dvc.lock
    metrics.json
nothing added to commit but untracked files present (use "git add" to track)
```

## A8. Reorganize: git mv & git rm

**[Terminal] A8. Reorganize with git mv and git rm**

```
$ git mv scratch_helper.py src/ && git rm src/scratch_helper.py
[dev 8f7e6ea] chore: remove obsolete helper using git rm
1 file changed, 1 deletion(-)
delete mode 100644 src/scratch_helper.py
```

## Part C — DVC Setup with Google Drive Remote

### C3. DVC Remote Configuration

```
[Terminal] C3. DVC Remote Configuration
$ dvc remote list
gdrive_storage  gdrive://1ALEBo2C0DXhzrc8mMcp5UK9XUAKd33WT      (default)
```

## Part D — DVC Pipeline Execution

### D3. Initial Pipeline Reproduction (v1)

[Terminal] D3. DVC Repro (v1 Baseline Pipeline Execution)

```
$ dvc repro
Running stage 'prepare':
w.python src/prepare.py
WARNING! All you messages before absl::InitializeLog() is called are written to STDERR
[2020-08-08 17:21:00.437531] 24864 port.cc:153] onednn custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPTS=0'.
C:\Users\LENDVO\AppData\Local\Temp\PythonPython313\site-packages\requests\_auth.py:111: RequestsDependencyWarning: urllib3 (2.5.0) or chardet (7.4.3)/charset-normalizer (3.4.2) doesn't match a supported version!
WARNING! All you messages before absl::InitializeLog() is called are written to STDERR
[2020-08-08 17:21:00.443506] 24864 port.cc:153] onednn custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPTS=0'.
Running stage 'preprocess':
w.python src/preprocess.py
Running stage 'train':
w.python src/train.py
WARNING! All you messages before absl::InitializeLog() is called are written to STDERR
[2020-08-08 17:21:00.500529] 24864 port.cc:153] onednn custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPTS=0'.
C:\Users\LENDVO\AppData\Local\Temp\PythonPython313\site-packages\requests\_auth.py:111: RequestsDependencyWarning: urllib3 (2.5.0) or chardet (7.4.3)/charset-normalizer (3.4.2) doesn't match a supported version!
WARNING! All you messages before absl::InitializeLog() is called are written to STDERR
[2020-08-08 17:21:00.605333] 24864 port.cc:153] onednn custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPTS=0'.
C:\Users\LENDVO\AppData\Local\Temp\PythonPython313\site-packages\keras\src\layers\reshaping\flatten.py:37: UserWarning: Do not pass an 'input_shape' / 'input_dim' argument to a layer. When using Sequential models, prefer using an 'Input(shape)' object as the first layer in the model instead.
super().__init__(**kwargs)
[2020-08-08 17:21:07.530030] 24864 cpu.feature_guard.cc:227] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
... [Truncated for display] ...
```

### D4. Hyperparameter Modification Reproduction (v2: dense\_units=256)

[Terminal] D4. DVC Repro with Hyperparameter Update (dense\_units= 256)

```
$ dvc repro
Stage 'prepare' didn't change, skipping
Stage 'preprocess' didn't change, skipping
Running stage 'train':
w.python src/train.py
WARNING! All you messages before absl::InitializeLog() is called are written to STDERR
[2020-08-08 17:21:00.640348] 14660 port.cc:153] onednn custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPTS=0'.
C:\Users\LENDVO\AppData\Local\Temp\PythonPython313\site-packages\requests\_auth.py:111: RequestsDependencyWarning: urllib3 (2.5.0) or chardet (7.4.3)/charset-normalizer (3.4.2) doesn't match a supported version!
WARNING! All you messages before absl::InitializeLog() is called are written to STDERR
[2020-08-08 17:21:00.791180] 14660 port.cc:153] onednn custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPTS=0'.
C:\Users\LENDVO\AppData\Local\Temp\PythonPython313\site-packages\keras\src\layers\reshaping\flatten.py:37: UserWarning: Do not pass an 'input_shape' / 'input_dim' argument to a layer. When using Sequential models, prefer using an 'Input(shape)' object as the first layer in the model instead.
super().__init__(**kwargs)
[2020-08-08 17:21:07.844680] 14660 cpu.feature_guard.cc:227] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
WARNING! tensorflow/tensorflow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.
Epoch 1/250
1s 17/250 [0m 37s] ----- [0m 1s] 32/0m 34/step - accuracy: 0.0156 - loss: 2.5527
1s 18/250 [0m 37s] ----- [0m 1s] 0m 0ms/step - accuracy: 0.2581 - loss: 2.0744
1s 19/250 [0m 37s] ----- [0m 1s] 0m 0ms/step - accuracy: 0.1300 - loss: 1.8600
1s 20/250 [0m 37s] ----- [0m 1s] 0m 0ms/step - accuracy: 0.1351 - loss: 1.7383
1s 21/250 [0m 37s] ----- [0m 1s] 0m 0ms/step - accuracy: 0.4384 - loss: 1.0340
1s 41/250 [0m 32s] 0m 17s ----- [0m 1s] 0m 7ms/step - accuracy: 0.4549 - loss: 1.5365
... [Truncated for display] ...
```

## Part E — Simulated Collaboration & Conflict Resolution

### E3. Git Merge Conflict in Code & dvc.lock

#### [Terminal] E3. Git Merge Conflict in Code & dvc.lock

```
$ git merge teammate-sim && git status
Auto-merging dvc.lock
CONFLICT (content): Merge conflict in dvc.lock
Auto-merging src/preprocess.py
CONFLICT (content): Merge conflict in src/preprocess.py
Automatic merge failed; fix conflicts and then commit the result.
On branch main
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)
Unmerged paths:
  (use "git add <file>..." to mark resolution)
    both modified:   dvc.lock
    both modified:   src/preprocess.py
Untracked files:
  (use "git add <file>..." to include in what will be committed)
  Assignments Report.pdf
assignment.pdf
no changes added to commit (use "git add" and/or "git commit -a")
```

### E4-E5. Conflict Resolved & Clean DVC Repo

#### [Terminal] E4-E5. Conflict Resolved & Clean DVC Repo

```
$ git commit -m "Merge branch teammate-sim..." && dvc status
[main 216a208] Merge branch 'teammate-sim': resolved code conflict in preprocess.py and synchronized dvc data
$ dvc status
Data and pipelines are up to date.
```

### Final Full Git Commit Graph

#### [Terminal] Final Git Repository Tree

```
$ git log --oneline --graph --all
+-----+
+ 216a208 Merge branch 'teammate-sim': resolved code conflict in preprocess.py and synchronized dvc data
+ |
+ | 2f09b6f feat(preprocess): teammate standardization normalization
+ | 3614e1e feat(preprocess): main branch min-max scaling to [-1, 1]
+ |
+ | 93792fc Merge dev branch (v2 pipeline) into main
+ |
+ | a4da01d experiment: tune dense units to 256 (v2)
+ | c1ace71 feat: record dvc pipeline v1 reproduction and lockfile
+ | 791eadb chore(dvc): initialize dvc and configure Google Drive remote
+ | b17e0ea chore: remove obsolete helper using git rm
+ | 6130abf refactor: move helper to src/ using git mv
+ | 5dc1d19 chore: add temporary scratch helper
+ | 4de9d4a feat(pipeline): add dvc.yaml stage definitions
+ | 017731e feat(avsl): add evaluate.py for test metrics and confusion matrix
+ | 91565dc feat(model): add train.py for Sequential ANN architecture
+ | 78dd057 feat(preprocess): add preprocess.py for normalization and validation split
+ | 9747683 feat(data): add prepare.py to download and extract Fashion-MNIST
+ | 3e9c23d feat(config): add params.yaml for centralized hyperparameters
+ |
+ | c49687d Merge branch hotfix into main
+ |
+ |... [truncated for display] ...
```